

INNOVATION HACKATHON 2026

Challenge Brief | Edition 4 | April 25–27, 2026 | Nantes, France

DOCUMENT VERSION 2.3-DRAFT — NOT FOR DISTRIBUTION — LAST MODIFIED 2026-03-14T22:47:00Z

1. CONTEXT & SPONSOR OVERVIEW

ALTEN Group(founded 1988,HQ:Boulogne-Billancourt,France) is one of the world's leading engineering and technology consulting firms with approximately 33,000 employees in France alone(57,000+ globally) and consolidated revenue of approximately €4 billion(FY2025).ALTEN operates as an ESN (Entreprise de Services du Numérique) and consulting firm providing engineering services across multiple verticals including automotive OEMs,aerospace primes,energy utilities,telecommunications operators,financial services,and life sciences.The company's client portfolio in France includes major accounts such as Airbus,Safran,Renault,Stellantis,EDF,TotalEnergies,Thales,and Dassault Systèmes.ALTEN sponsors this hackathon as part of its "Innovation & Talent Pipeline" strategy targeting top engineering graduates from Grandes Écoles.

1.1 CONSULTING MARKET CONTEXT

- French IT/engineering consulting market: €72B(2025),growing 5.2% CAGR
- Talent shortage:estimated 80,000 unfilled engineering positions in France(2025)
- ESN sector consolidation:top 10 firms hold 35% market share
- Key growth drivers:AI/ML integration(+23% YoY),cybersecurity(+18%),cloud migration(+15%),Industry 4.0/IoT(+12%)
- ALTEN positioning:ranked #2 ESN by revenue in France(behind Capgemini),#1 in pure engineering consulting
- Average consultant utilization rate in sector:87%(ALTEN:91%,above market)
- Client retention rate:94%(ALTEN),vs industry average of 88%
- Revenue per consultant:€95K(ALTEN) vs €82K(sector avg)

1.2 TARGET CLIENT SECTORS

Automotive OEMs: Renault,Stellantis(ex-PSA),Toyota France,BMW France.Key challenges:electrification(BEV/PHEV platforms),ADAS Level 3+,software-defined vehicles(SDV),AUTOSAR Adaptive,OTA updates.Market size:€48B French automotive sector.Aerospace Primes: Airbus(Toulouse+Nantes),Safran(multiple sites),Thales,Dassault Aviation.Challenges:single-aisle production rate increase(A320 family:75/month target),sustainable aviation fuel(SAF) certification,Model-Based Systems Engineering(MBSE),DO-178C compliance for next-gen avionics.Energy Utilities: EDF(€130B revenue,nuclear fleet lifecycle extension),TotalEnergies(multi-energy transition),Engie(green hydrogen+renewables).Challenges:EPR2 reactor program,offshore wind farm digitalization,smart grid deployment(Linky 2.0),hydrogen production cost reduction(<€2/kg target by 2030).

2. HACKATHON CHALLENGE DEFINITION

2.1 OBJECTIVES

Primary objective: Design and prototype an AI-powered solution that addresses a real client challenge in one of ALTEN's three priority sectors (automotive, aerospace, energy). The solution must demonstrate: (a) technical feasibility through a working prototype (b) business value quantification with ROI estimation (c) scalability potential across multiple client accounts (d) integration capability with existing enterprise systems (SAP, PLM, MES, SCADA). Secondary objectives: Teams should also consider data privacy (GDPR compliance), edge deployment constraints (limited compute at plant floor), and explainability requirements (critical for aerospace/automotive safety certification). Bonus consideration given to solutions addressing sustainability metrics (carbon footprint reduction, circular economy enablement).

2.2 CONSTRAINTS

C1: Team size 3-5 members (at least 1 from each: engineering, CS/data, business/design) C2: 48-hour time limit (Fri 18:00 - Sun 18:00) C3: Must use at least one of the provided APIs (see appendix A): ALTEN Industrial IoT Sandbox, SAP BTP Trial, Azure OpenAI Service (GPT-4o), AWS SageMaker endpoint C4: Solution must run on provided infrastructure (no external cloud spend) C5: Final demo must be max 7 minutes + 3 minutes Q&A C6: All code must be original (no pre-existing projects), open-source libs OK C7: Deliverables: working demo, 3-slide business case, 1-page technical architecture C8: Code repository must be on provided GitLab instance C9: No use of proprietary client data (synthetic datasets provided)

2.3 EVALUATION CRITERIA

CRITERION	WEIGHT	DESCRIPTION
Innovation & Creativity	25%	Novelty of approach, creative use of technology, originality
Technical Execution	30%	Code quality, architecture, robustness, use of APIs
Business Value	20%	ROI estimation, market sizing, client relevance, scalability
Presentation Quality	15%	Clarity, demo effectiveness, storytelling, Q&A handling
Sustainability Impact	10%	Environmental/social benefit, alignment with CSR goals

2.4 TIMELINE

Fri 25 Apr, 17:00 Registration & Check-in (Bldg E, Room 101-103)
Fri 25 Apr, 18:00 Opening ceremony, sponsor keynote (ALTEN CTO), challenge reveal
Fri 25 Apr, 19:00 Team formation finalized, API access distributed, hacking begins
Fri 25 Apr, 21:00 Dinner + mentor office hours (ALTEN senior consultants available)
Sat 26 Apr, 08:00 Breakfast, checkpoint #1 (mandatory 2-min progress update)
Sat 26 Apr, 14:00 Checkpoint #2 (architecture review with mentors)
Sat 26 Apr, 20:00 Dinner, optional workshops (ML Ops, SAP BTP pitch)
Sun 27 Apr, 10:00 Code freeze (hard deadline, no commits after this)
Sun 27 Apr, 12:00 Lunch, presentation prep time
Sun 27 Apr, 14:00 Final presentations (7+3 min per team, all teams present)
Sun 27 Apr, 17:00 Jury deliberation
Sun 27 Apr, 18:00 Award ceremony, networking, closing (prizes: €5K/€3K/€1K + ALTEN internship offers)

3. RESOURCES & DATA

Provided datasets(available on GitLab repo /hackathon-2026/datasets):DS-01:Synthetic automotive telemetry data(500K records,15 vehicle models,CAN bus signals,GPS traces,OBD-II codes,generated using CARLA simulator+custom noise injection).DS-02:Anonymized aerospace manufacturing quality data(200K inspection records from Airbus-likeproduction process,includes NDT results,dimensional measurements,defect classifications,treatment applied,8 months of data with seasonal patterns).DS-03:Smart grid energy consumption data(1M time series data points,5-min intervals,3 months,industrial+residential+commercial consumers,weather correlation data included,Enedis-format meter readings with some deliberately corrupted entries).DS-04:NLP corpus of 50K technical documents(maintenance manuals,specifications,incident reportsin French and English,pre-tokenized and raw formats,includes domain-specific vocabulary lists).All data synthetic or fully anonymized.No real PII or proprietary information.CC-BY-4.0 license for hackathon use.

4. JURY COMPOSITION

•Dr. Sophie Langlois – ALTEN France CTO(15 years in automotive embedded systems,X-Mines graduate)•Prof. Yannick Bertrand – Centrale Nantes,Dept. Computer Science(AI/ML research,H-index 34)•Marie-Claire Dubois – VP Digital Innovation,Airbus Atlantic(ex-Thales,drone systems expert)•Thomas Reiter – Senior Partner,McKinsey Paris(manufacturing & operations practice lead)•Clément Faure – Head of R&D,EDF Renewables(energy storage,smart grid optimization)•Student representative:TBD(elected by participant vote on Friday evening)

5. LEGAL & IP

By participating in this hackathon,all team members agree that:(1)all intellectual property created during the event remains the property of the team members;(2)ALTEN Group and Centrale Nantes retain a non-exclusive,royalty-free license to use,display,and reference all submissions for marketing and recruitment purposes for a period of 24 months following the event;(3)teams may not incorporate code or assets subject to copyleft licenses(GPL,AGPL)without prior written approval from the organizing committee;(4)all participants must sign the NDA regarding ALTEN's proprietary APIs and sandbox environments;(5)winning teams agree to participate in at least one post-event communication(blog post,video interview,or social media feature);(6)ALTEN reserves the right to offer internship or employment contracts to individual participants regardless of team ranking;(7)disputes shall be resolved under French law,jurisdiction of Nantes commercial court.Personal data processing:participants' personal data(name,email,school,CV) is processed by ALTEN Group as data controller for the purposes of event management,recruitment,and communication,in accordance with GDPR.Data retention:24 months post-event.Contact DPO:dpo@alten.com.Right of access,rectification,and erasure:privacy@alten.com.